UBER Performance Analysis

1. Project Overview

This project analyzes ride-booking operations using transactional data from a dataset ofUber bookings on India's leading ride-hailing platform. The goal is to uncover insights into booking completion trends, cancellation patterns, revenue performance, and the impact of vehicle type, and ride distance on customer and driver experience to guide strategic operational decisions..

2. Dataset Summary

- Rows: 70,000 — Columns: 20

- Key Features:
- Customer details (Customer ID, Customer Rating)

- Vehicle details (Vehicle_Type)

- Booking details (Booking ID, Booking Status, Pickup Location, Drop Location, Date, Time, Payment Method, Booking Value, Ride Distance)

- Operational metrics (V_TAT, C_TAT, Driver Ratings, Incomplete Rides Reason)

- Cancellation flags (Canceled Rides by Customer, Canceled Rides by Driver, Incomplete Rides)

  Missing Data: Null values in Incomplete_Rides_Reason, Canceled_Rides_by_Customer, and Canceled_Rides_by_Driver columns (expected — only present for rides that were actually cancelled or incomplete)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

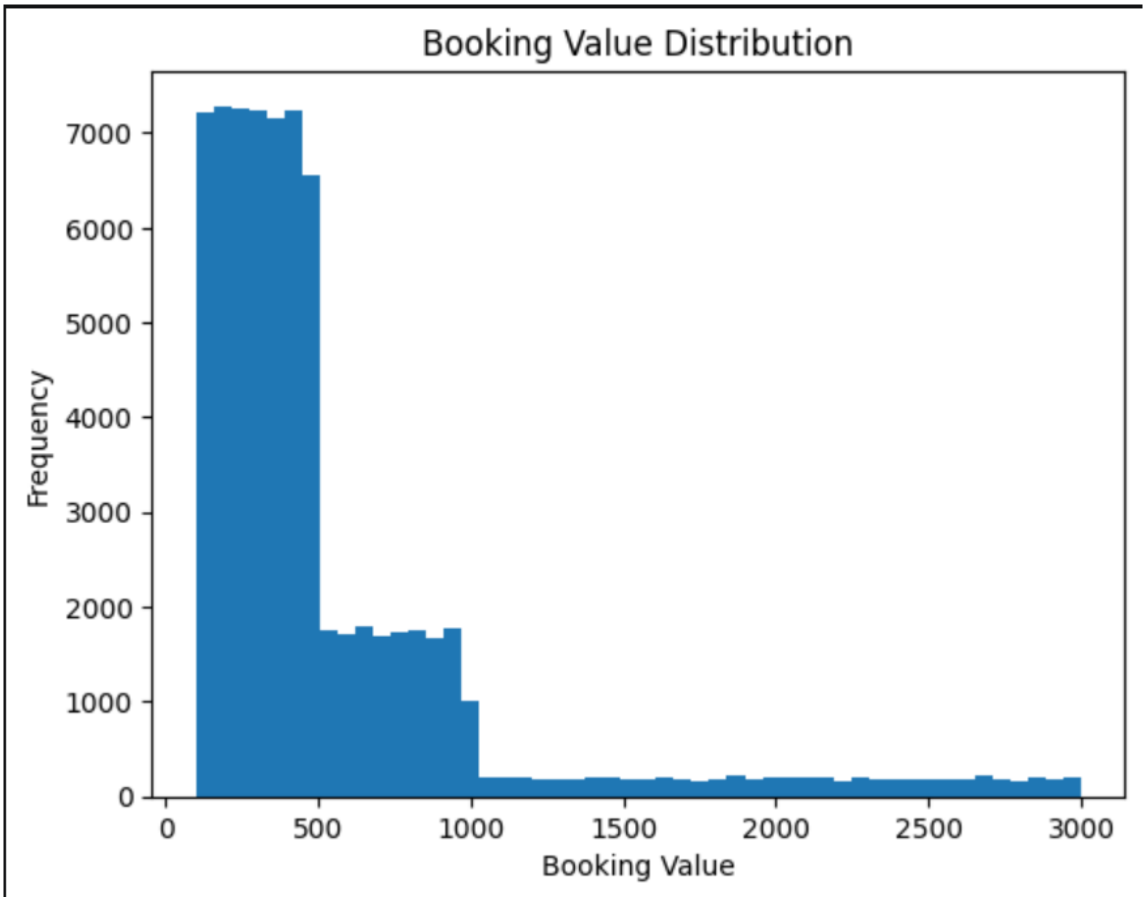
- **Data Loading:** Imported the dataset using `pandas`.

- **Initial Exploration:** Used `.describe()` for summary statistics and `.median()` to know the where the spread of the data is present

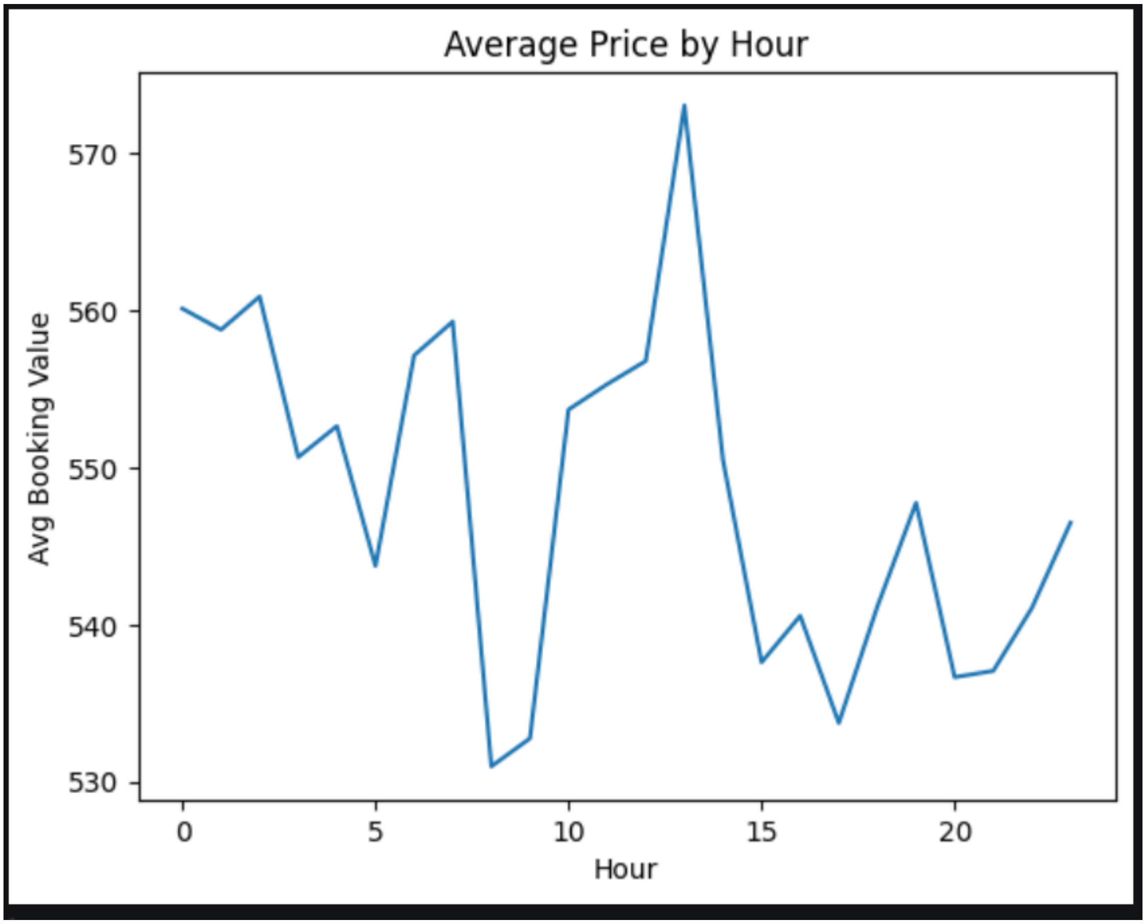
- **Missing Data Handling:** Checked for null values and imputed missing values.

	count	mean	std	min	25%	50%	75%	max	medain	IQR
V_TAT	44271.0	171.323620	80.682293	35.0	98.0	168.0	238.0	308.0	168.0	140.0
C_TAT	44271.0	84.955275	35.959016	25.0	55.0	85.0	115.0	145.0	85.0	60.0
Booking_Value	71201.0	548.233901	535.399324	100.0	242.0	385.0	622.0	2999.0	385.0	380.0

- **Explored Booking Value Distribution**



- **Average Booking Value Peaks By Hours**



- **Data Consistency Check:** Verified that Canceled_Rides_by_Customer, Canceled_Rides_by_Driver, and Incomplete_Rides_Reason columns contained null values only for successfully completed rides confirming these nulls are structurally valid and not data quality issues, so no imputation was required..

- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

Performed structured analysis in MySQL to answer key business questions:

1. Booking Status Breakdown – Compared Rides That were Successful, Incomplete or Cancelled

Booking_Status	total_bookings
Success	19946
Canceled by Driver	5729
Canceled by Customer	3302
Driver Not Found	3197

2. **Revenue by vehicle type** – Calculated which Vehicle Type generates the most Revenue

Vehicle_Type	total_revenue
Prime Sedan	1643230
Bike	1568017
eBike	1558869
Auto	1545974
Prime SUV	1545735
Mini	1521902
Prime Plus	1517256

3. **Top Pickup Locations** – Found out the Top 5 Pickup Locations and their total bookings

Pickup_Location	total_bookings
Indiranagar	697
Kammanahalli	694
Banashankari	686
Sarjapur Road	682
Ramamurthy Nagar	681

4. **Average Customer and Driver Rating by Vehicle Type** – Calculated the Average Rating delivered by the Customer and the Driver for each Vehicle Type

Vehicle_Type	avg_driver_rating	avg_customer_rating
eBike	4.02	3.99
Prime Plus	4.01	4
Prime SUV	4	3.99
Auto	4	4
Mini	3.99	4.01
Prime Sedan	3.99	4
Bike	3.98	3.98

5. **Incomplete Ride Reasons** – Found the Most common reasons why the rides go incomplete

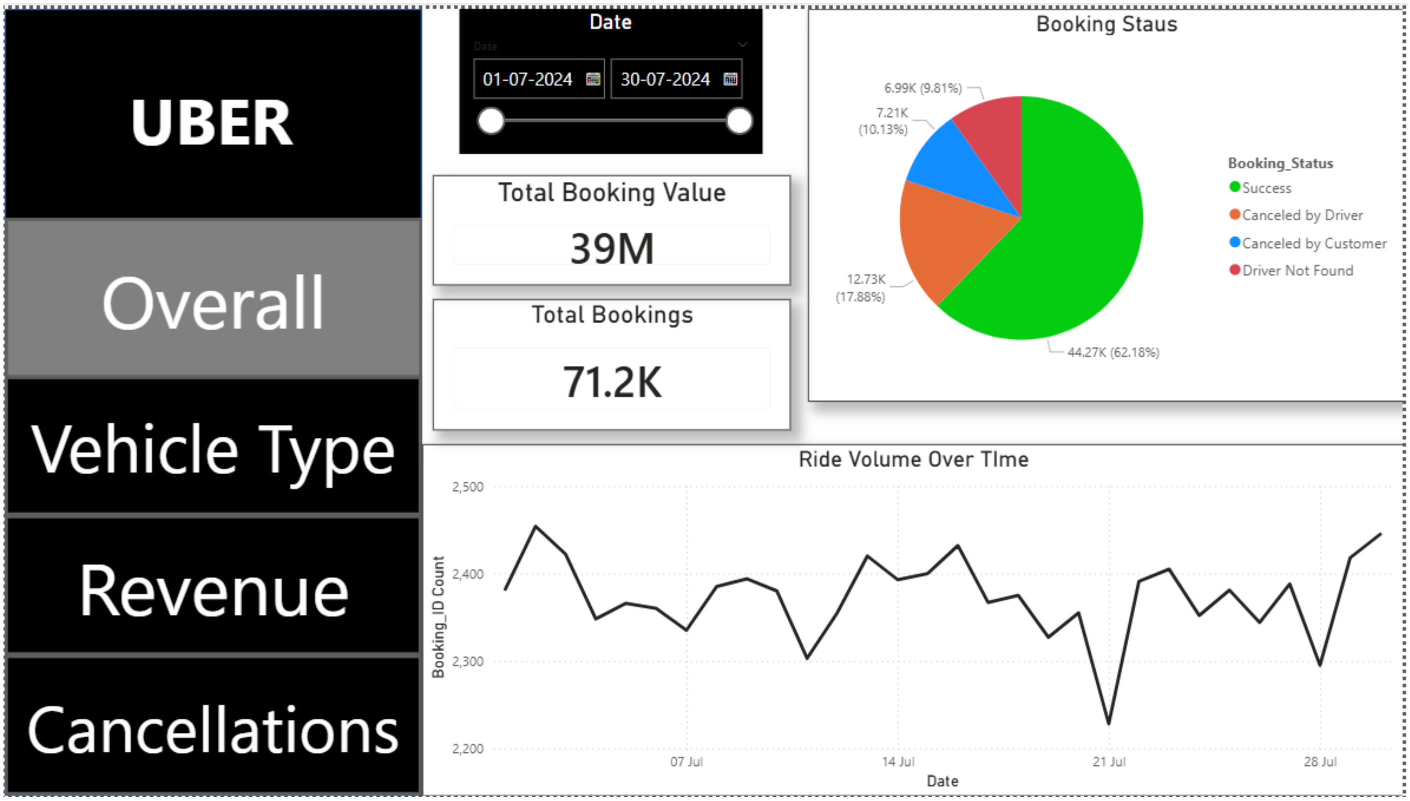
Incomplete_Rides_Reason	total_count
Vehicle Breakdown	500
Customer Demand	483
Other Issue	244

6. **Avg TAT** – Calculated the Averae TAT for successful boookings

Booking_Status	avg_vehicle_tat	avg_customer_tat
Success	170.62	85.21

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



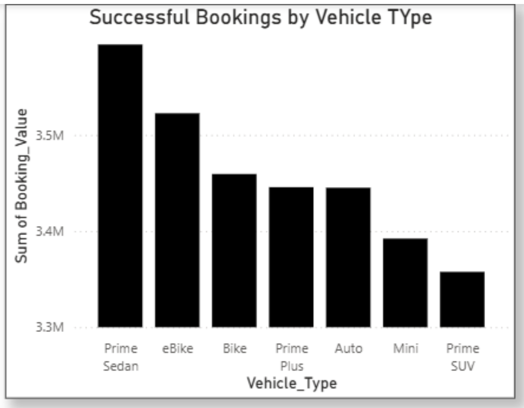
UBER

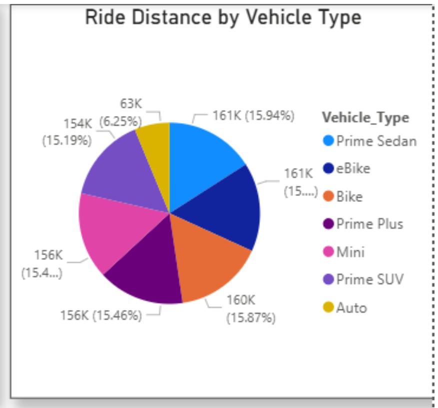
Overall

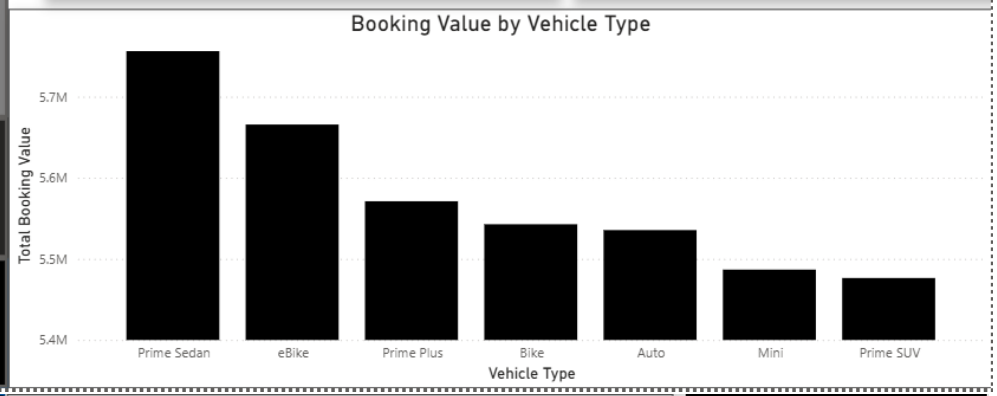
Vehicle Type

Revenue

Cancellations







UBER

Overall

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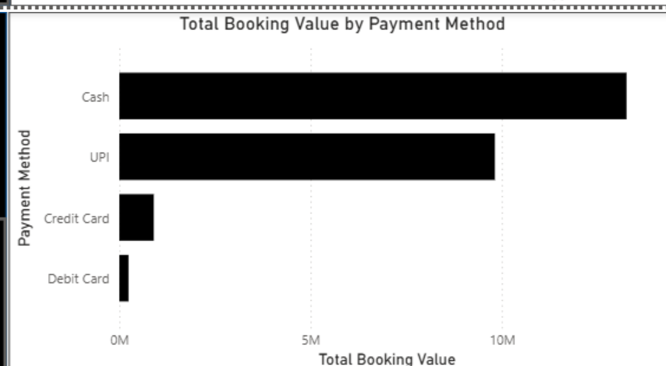
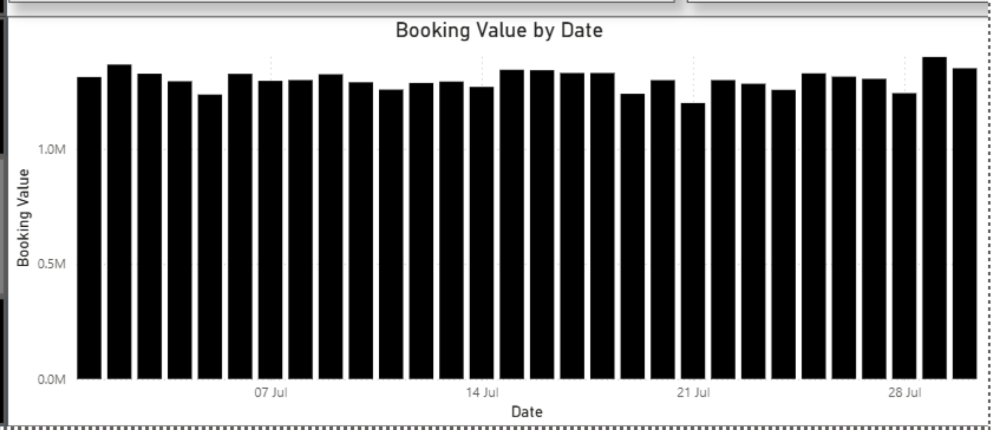


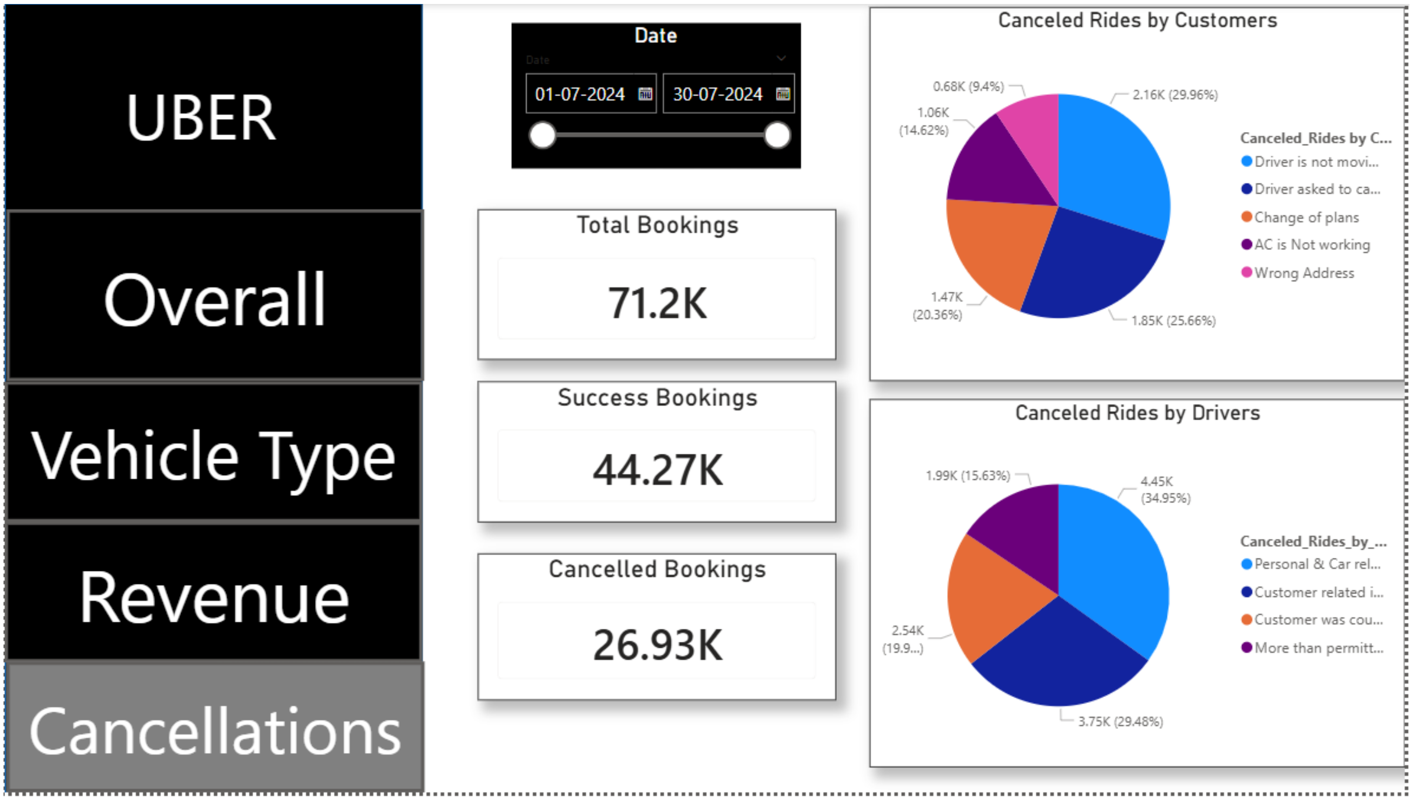
Date

01-07-2024 30-07-2024

Top 5 Customers

Customer_ID	Sum of Booking_Value
CID635848	4976
CID792698	4457
CID825950	4729
CID889014	4685
CID933539	5314
Total	**24161**




UBER
Overall
Vehicle Type
Revenue
Cancellations
Date
01-07-2024
30-07-2024
Total Bookings
71.2K
Success Bookings
44.27K
Cancelled Bookings
26.93K
Canceled Rides by Customers
0.68K (9.4%)
2.16K (29.96%)
1.06K (14.62%)
1.47K (20.36%)
1.85K (25.66%)
Canceled_Rides by C...
Driver is not movi...
Driver asked to ca...
Change of plans
AC is Not working
Wrong Address
Canceled Rides by Drivers
1.99K (15.63%)
4.45K (34.95%)
2.54K (19.9...)
3.75K (29.48%)
Canceled_Rides_by_...
Personal & Car rel...
Customer related i...
Customer was cou...
More than permitt...

6. Business Recommendations

- **Reduce Cancellations by Driver:** Driver initiated cancellations directly impact platform reliability and customer trust. Uber should introduce penalty systems or incentive-based rewards for drivers who maintain low cancellation rates, and improve driver-customer matching algorithms to reduce mismatches that lead to cancellations.

- **Optimise High Revenue Vehicle Types :** Certain vehicle categories generate significantly more revenue per ride. Uber should prioritise driver recruitment and retention in these categories and offer loyalty pricing or subscription plans to frequent users of high-value vehicle types to further boost revenue..

- **Reduce Incomplete Rides** : Incomplete rides represent direct revenue leakage. The most common reasons should be addressed operationally whether through better route guidance, vehicle maintenance checks, or clearer driver protocols to push the booking completion rate above the platform benchmark.

- **Improve TAT for Better Experience :** With average vehicle TAT at 170 seconds, reducing wait times even by 20-30 seconds can meaningfully improve customer ratings. Deploying more drivers in high-demand pickup zones during peak hours is the most direct lever to achieve this.%.